WEAKLY ASSOCIATED POINTS OF A QUASI-COHERENT MODULE ARE DETECTED AFTER AN ETALE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

Editorial note. The same-chapter referenced result is located at its \label line inside the pinned chapter file; the cross-chapter standard-etale lemma is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Weakly associated points of a quasi-coherent module. Introduced earlier in the same project (Divisors, Definition divisors-definition-weakly-associated) and imported by this chapter. Supplies the notion WeakAss appearing on both sides of the asserted equivalence; the reduction that transports it through the etale morphism stays inside the counted proof span.
- Reduction of an etale morphism to a standard etale morphism. Morphisms, Lemma labelled morphisms-lemma-etale-locally-standard-etale (canonical Stacks tag 02GT); invoked at proof line 367. Supplies the local normal form S = Spec(A), U = Spec(A[x, 1/g]/(f)) with f monic and f' invertible; the identification of that algebra as a localisation of a finite locally free A-algebra is the counted step.

Source reference. Original lemma 05FN: sources/stacks-flat/pinned-original-source.tex, source lines 312–312. Same-chapter pointer invoked at proof line 375 for the finite locally free case; the two counted steps that put U inside a finite locally free S-scheme (localisation of A[x]/(f) and the standard etale normal form) are retained verbatim in the delivered body.

Source reference. Original lemma 02GT: sources/stacks-chow/tags.txt, source lines 3240–3240. Cross-chapter standard prerequisite invoked once at proof line 367 for the local normal form of an etale morphism; it supplies the shape S = Spec(A), X = Spec(A[x, 1/g]/(f)) only, and the finite-free-envelope argument is retained in the counted span.

Lemma 0.1. *Let $h : U \to S$ be an étale morphism of schemes. Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_S$-module. Let $u \in U$ be a point with image $s \in S$. Then*

$$u \in \mathit{WeakAss}(h^*\mathcal{G}) \Leftrightarrow s \in \mathit{WeakAss}(\mathcal{G})$$

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Proof. After replacing S and U by affine neighbourhoods of s and u we may assume that g is a standard étale morphism of affines, see Morphisms, Lemma 02GT. Thus we may assume $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(A[x, 1/g]/(f))$, where f is monic and f' is invertible in $A[x, 1/g]$. Note that $A[x, 1/g]/(f) = (A[x]/(f))_g$ is also the localization of the finite free A-algebra $A[x]/(f)$. Hence we may think of U as an open subscheme of the scheme $T = \mathrm{Spec}(A[x]/(f))$ which is finite locally free over S. This reduces us to Lemma 05FN above. $\square$